# The Effects of Colorado's Pay Transparency Law on Older Workers

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# Introduction: Pay Transparency and Older Workers

- **The problem:** Pay disparities and discrimination persist in the labor market.
  - Wage information frictions and workers' misperceptions may strengthen employer wage-setting power and contribute to pay disparities.
- **One policy response:** Mandatory pay disclosure in job postings
  $\Rightarrow$ Workers can compare pay across firms and update their beliefs about outside wages.
- **Why older workers?** Less up-to-date wage information and more discrimination.
  $\Rightarrow$ Pay disclosure may be more valuable for older workers.

**Research question: How does cross-firm pay transparency affect the labor market outcomes of older workers?**

# Motivation: Why Pay Transparency May Matter

**Sources of wage misperceptions:** Workers anchor wage expectations on current or previous wages and update them only gradually (Krueger and Mueller, 2016; Mueller et al., 2021; Jäger et al., 2024).

**What pay transparency can do:**

**Wage information in job postings** $\xrightarrow{\text{update}}$ **Beliefs about outside opportunities** $\xrightarrow{\text{change}}$ **Reservation wages and bargaining Job search and mobility**

*Broader market effect:* May also strengthen wage competition across firms (Arnold et al., 2025).

# Motivation: Why Pay Transparency May Matter for Older Workers

**Cross-firm pay transparency may have particularly important effects on mid- to late-career workers.**

- **Information gaps:** Less current information about outside wages.
- **Hiring frictions:** Discrimination and longer job searches (Neumark et al., 2019; Brown et al., 2025).
- **Retirement margin:** Labor force exit as an alternative response (Maestas and Li, 2006; Blundell et al., 2016).

**Broader policy relevance:** As populations age, older workers' employment has important implications for tax revenues and public spending.

# Empirical Setting and Research Design

**Method:** Difference-in-differences using variation across states and over time.

- **Policy setting:** Colorado's Equal Pay for Equal Work Act (EPEWA) took effect on January 1, 2021.
- **Treated and comparison states:** Colorado is the treated state. Comparison states have salary history bans but no cross-firm pay transparency laws through 2025.
- **Estimated effect:** The coefficient captures the post-EPEWA change in labor market outcomes in CO relative to the contemporaneous change in the comparison states.
- **Identifying assumption:** Without EPEWA, labor market outcomes in Colorado would have followed the same trends as those in the comparison states.

# Main Results

Pay transparency **improves wage outcomes** for mid- to late-career workers:

- Weekly earnings increase by **4.6%** and hourly wages by **3.7%**.
- Wage gains strengthen over time.

The gains are concentrated among **women and highly educated workers** and larger among **the group near the retirement age**.

- The gender earnings gap falls by **6.7%** and the gender wage gap by **3.7%**.

Pay transparency also increases **employment (0.6 p.p.) and working hours (0.603 hours)**, but these effects are concentrated in the short run.

Effects are **limited among younger workers aged 25–44 overall**.

# Main Results: Heterogeneity and Mechanisms

**Job mobility:** Pay transparency increases hourly wages and reduces the gender wage gap among both job changers and job stayers, with larger effects among **job changers**.

**Occupational heterogeneity:** Effects vary with baseline wage levels, wage dispersion, age-wage profiles, and local labor market concentration (occupation-by-metropolitan-area HHI).

- Larger wage gains in occupations with **greater wage dispersion**.
- Largest wage gains in **medium-wage occupations**.
- Larger wage gains in more concentrated local labor markets (higher HHI).
- Larger gender wage gap reductions in occupations with **steep age–wage profiles**.

**Workers' beliefs:** Following pay transparency, mid- to late-career workers:

- Increase expected wages from outside opportunities and their reservation wages;
- Temporarily become more optimistic about receiving a job offer.

# Related Literature and Contributions

**Effects of pay transparency:** Card et al. (2012); Mas (2017); Bennedsen et al. (2022); Baker et al. (2023); Cullen and Pakzad-Hurson (2023); Skoda (2022); Feng (2024); Arnold et al. (2025); He and Upmanyu (2026).

**Contribution:** Shows larger gains from reducing wage-information frictions among mid- to late-career workers, who face greater information and hiring barriers.

**Labor markets of mid- to late-career workers:** Chan and Stevens (2001); Coile et al. (2014); Anri (2015); Brown et al. (2025); Lahey (2008); Neumark et al. (2019); Burn et al. (2019); Homrighausen and Lang (2019).

**Contribution:** Provides policy evidence that reducing wage-information frictions can partially mitigate older workers' labor-market disadvantages.

**Worker beliefs about outside opportunities:** Krueger and Mueller (2016); Mueller et al. (2021); Jäger et al. (2024); Caldwell and Harmon (2019); Altmann et al. (2020); Roussille (2021).

**Contribution:** Provides complementary evidence linking pay transparency, belief updating, and labor-market outcomes.

# Policy Background: Colorado EPEWA

**Colorado's Equal Pay for Equal Work Act (EPEWA)**
Effective January 1, 2021

**Main provision: Pay disclosure**

Employers disclose compensation and benefits information in job postings (a good-faith assessment of the position).

**Other provisions**

- Promotion notices
- Recordkeeping requirements
- Salary-history restrictions

**Coverage:** All firms, regardless of size. Remote jobs are covered when employers have staff in Colorado.

# Policy Background: Enforcement

**Enforcement agency: Colorado Department of Labor and Employment (CDLE)**, Colorado's primary labor regulator.

↓

**Complaint-based, education-first procedure**

Individuals can report noncompliant postings.

Employers are generally allowed to correct violations.

↓

**Penalties:** Fines range from \$500 to \$10,000 per posting.

↓

As of July 7, 2026: 2,860 complaints → 632 voluntary compliance letters → 201 formal investigations → 24 citations. 78.55% were corrected before formal investigation; 10 of 24 citations included monetary penalties.

▸ Policy Enforcement Appendix

# Background: Salary History Bans (SHBs)

**What are SHBs?**

Employers cannot ask about or use an applicant's past pay when setting compensation for a new job.

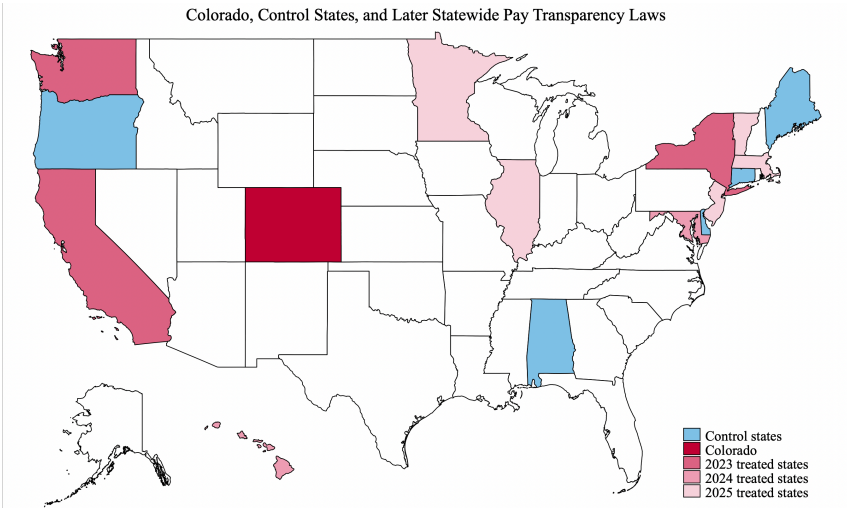
**Policy motivation**

- Prevent past wage discrimination from carrying over into new jobs
- Base pay on qualifications and job requirements rather than prior earnings

# Background: Statewide Policy Implementation Table

| Cross-Firm Pay Transparency | | Salary History Ban | |
|---|---|---|---|
| **Location** | **Effective Year** | **Location** | **Effective Year** |
| Colorado | 2021 | Delaware | 2017 |
| California | 2023 | New York | 2017 |
| Washington | 2023 | Oregon | 2017 |
| New York | 2023 | California | 2018 |
| Hawaii | 2024 | Massachusetts | 2018 |
| Washington, D.C. | 2024 | New Jersey | 2018 |
| Maryland | 2024 | Vermont | 2018 |
| Illinois | 2025 | Alabama | 2019 |
| Minnesota | 2025 | Connecticut | 2019 |
| New Jersey | 2025 | Illinois | 2019 |
| Massachusetts | 2025 | Maine | 2019 |
| Vermont | 2025 | Washington | 2019 |
| | | Maryland | 2020 |
| | | Colorado | 2021 |
| | | Nevada | 2021 |
| | | Rhode Island | 2023 |

# Background: Policy Implementation Map



# Data: IPUMS-CPS & ORG Supplements

- Monthly IPUMS-CPS, 2014–2025 (Flood et al., 2023).
- CPS follows a 4-8-4 rotation design:
  - interviewed for 4 consecutive months,
  - out of sample for 8 months,
  - then interviewed for another 4 months.
- Main variables:
  - Demographics: age, sex, marital status, race, education, family size, location;
  - Labor market outcomes: employment, working hours, class of worker, industry, occupation, whether changed employers.
- ORG supplement: earnings information in the 4th and 8th interview months:
  - hourly wage,
  - weekly earnings.

# Data: Sample Selection

- Sample: workers aged 45–64 from 2014–2025.
- Treated state: CO.
- Control group: AL, CT, DE, ME, and OR (Choosing other SHB states refer to Feng (2025) and Cullen (2024)).
- Restrictions: individuals with citizenship status who reside in metropolitan areas and are not self-employed.

▸ Why CO ▸ Why 45-64

# Empirical specification: Wages and Labor Supply

$$Y_{igt} = \beta \, \text{CO}_{ig} \times Post_t + X'_{igt}\gamma + \lambda M_{st} + \omega_{oy(t)} + \mu_{jy(t)} + \delta_t + \theta_g + \varepsilon_{igt}, \qquad (1)$$

where $i$ indexes individuals, $s$ indexes states, $g$ indexes metropolitan areas, $o$ indexes occupations, $j$ indexes industries, $y(t)$ is calendar year, and $t$ is calendar month (defined as year-by-month cells).

- $\omega_{oy(t)}$: occupation-by-year fixed effects;
- $\mu_{jy(t)}$: industry-by-year fixed effects;
- $\delta_t$: year-month fixed effects;
- $\theta_g$: metropolitan area fixed effects;

ES Main

# Empirical specification: Gender Wage Gap

$$Y_{igt} = \beta_1^G \, \text{CO}_{ig} \times Post_t + \beta_2^G \, \text{CO}_{ig} \times Post_t \times Female_i + \phi \, Post_t \times Female_i + X'_{igt}\gamma + \lambda M_{st} + \omega_{oy(t)} + \mu_{jy(t)} + \delta_t + \theta_g + \eta_{fg} + \varepsilon_{igt}, \quad (2)$$

- $\eta_{fg}$: female-by-metropolitan-area fixed effects;
- $\beta_2^G$ measures the marginal effect of pay transparency on women's wage relative to men's wage. A positive $\beta_2^G$ implies a narrowing of the gender gap.

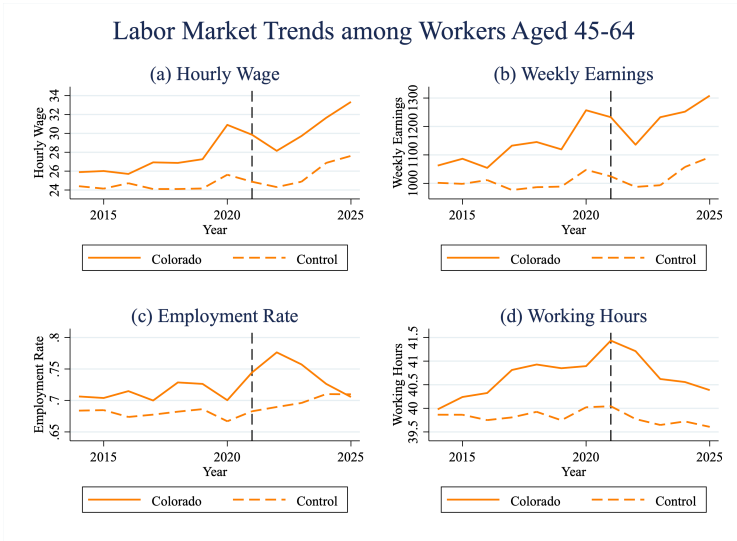
ES Gap

# Data: Summary Statistics

| Variable | Colorado | | Other SHB States | | Difference |
|---|---|---|---|---|---|
| | Mean | Std. Dev. | Mean | Std. Dev. | |
| Age | 54.541 | 5.800 | 54.814 | 5.703 | -0.273† |
| Sex (Male=1) | 0.431 | 0.495 | 0.421 | 0.494 | 0.010† |
| Marriage | 0.670 | 0.470 | 0.644 | 0.479 | 0.026† |
| Less than high school | 0.045 | 0.206 | 0.063 | 0.243 | -0.018† |
| High school | 0.209 | 0.406 | 0.293 | 0.455 | -0.084† |
| Some college | 0.146 | 0.353 | 0.154 | 0.361 | -0.008† |
| Associate degree | 0.101 | 0.302 | 0.108 | 0.311 | -0.007† |
| Bachelor degree | 0.301 | 0.459 | 0.231 | 0.421 | 0.070† |
| Master or more | 0.198 | 0.399 | 0.150 | 0.357 | 0.048† |
| White | 0.799 | 0.400 | 0.781 | 0.414 | 0.018† |
| Black | 0.035 | 0.184 | 0.128 | 0.335 | -0.093† |
| Hispanic | 0.117 | 0.322 | 0.047 | 0.213 | 0.070† |
| Family size | 2.618 | 1.325 | 2.577 | 1.279 | 0.041† |
| Num of children in HH | 1.379 | 2.012 | 1.248 | 1.878 | 0.131† |
| Num of own children in HH | 0.734 | 1.023 | 0.670 | 0.965 | 0.064† |
| Paid hourly | 0.102 | 0.302 | 0.125 | 0.330 | -0.023† |
| Public sector | 0.199 | 0.399 | 0.185 | 0.389 | 0.014† |
| Employment status | 0.722 | 0.448 | 0.686 | 0.464 | 0.036† |
| Main-job hours if employed | 40.847 | 9.760 | 40.025 | 9.042 | 0.822† |
| Weekly earnings if employed | 1,154.462 | 692.824 | 1,012.773 | 633.321 | 141.689† |
| Hourly wage if employed | 28.068 | 15.906 | 24.869 | 14.514 | 3.199† |
| Observations (main vars) | 33,366 | | 185,994 | | |
| Observations (main-job hours) | 23,191 | | 119,983 | | |
| Observations (weekly earnings) | 6,164 | | 32,247 | | |
| Observations (hourly wage) | 5,928 | | 30,400 | | |

*Notes:* † indicates significance at the 1 percent level, and ‡ indicates significance at the 5 percent level.

# Data: Labor Market Trends by Age Group



# DiD Estimates: Weekly Earnings & Hourly Wage

| | ln(Weekly earnings) | | | ln(Hourly wage) | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| CO × Post | 0.046*** | 0.042*** | 0.048*** | 0.037*** | 0.033*** | 0.043*** |
| | (0.016) | (0.014) | (0.013) | (0.012) | (0.010) | (0.010) |
| Metropolitan Area FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | No | Yes | Yes | No | Yes | Yes |
| Month FE | No | Yes | Yes | No | Yes | Yes |
| Year × Month FE | Yes | No | No | Yes | No | No |
| Industry × Year FE | Yes | Yes | No | Yes | Yes | No |
| Occupation × Year FE | Yes | Yes | No | Yes | Yes | No |
| Observations | 38,411 | 38,411 | 38,411 | 36,328 | 36,328 | 36,328 |
| Adjusted $R^2$ | 0.358 | 0.357 | 0.279 | 0.420 | 0.419 | 0.340 |

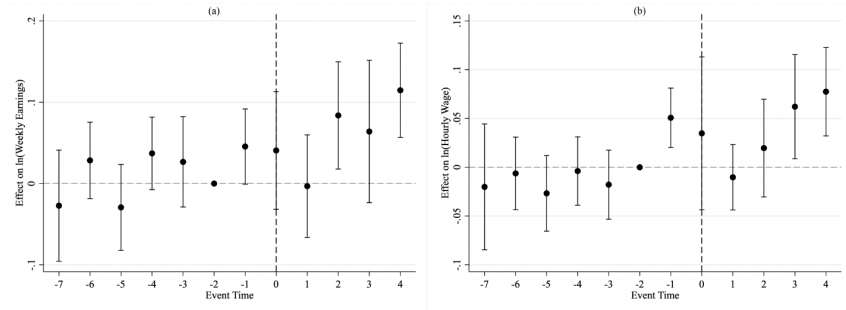
Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Hetero Hourly Paid ▸ Robust DDD Wage ▸ Robust DDD Gap

- Increases weekly earnings by 4.6% ($\approx$ \$46.59/week).
- Increases hourly wage by 3.7% ($\approx$ \$0.92/hour).

# Event Study: Weekly Earnings & Hourly Wage




- Both effects were limited initially but gradually increased over time.
- Pre-policy estimates are generally flat, except in 2020, likely due to differential COVID-19 wage shocks. Choose event time $-2$ as the omitted period (Alstadsæter et al., 2017; Kleven et al., 2025).

▸ EsAltBase ▸ DigWageCOVID

# Heterogeneous Effects Based on Gender & Education

| | Male | Female | Low educated | High educated |
|---|---|---|---|---|
| *Panel A: ln(Weekly earnings)* | | | | |
| CO×Post | 0.011 | 0.082*** | -0.019 | 0.067*** |
| | (0.022) | (0.020) | (0.023) | (0.012) |
| Observations | 17,581 | 20,823 | 11,081 | 27,329 |
| Adjusted $R^2$ | 0.329 | 0.325 | 0.258 | 0.318 |
| *Panel B: ln(Hourly wage)* | | | | |
| CO×Post | 0.019 | 0.057*** | 0.002 | 0.050*** |
| | (0.018) | (0.013) | (0.019) | (0.012) |
| Observations | 16,611 | 19,711 | 10,378 | 25,949 |
| Adjusted $R^2$ | 0.404 | 0.401 | 0.306 | 0.342 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- **Gender:** Women experience substantially larger earnings and wage gains than men, driving the reduction in gender disparities.
- **Education:** Highly educated workers experience substantially larger earnings and wage gains than less-educated workers.

# DiD Estimates: Gender Earning/Wage Gap

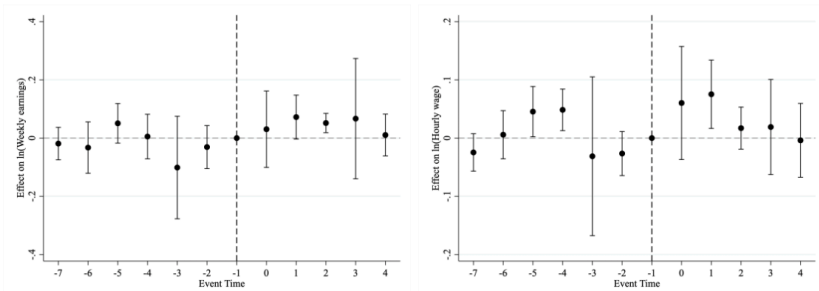
| | ln(Weekly earnings) | | ln(Hourly wage) | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| CO × Post | 0.009 | 0.013 | 0.019 | 0.017 |
| | (0.018) | (0.019) | (0.018) | (0.016) |
| CO × Post × Female | 0.066*** | 0.067*** | 0.037** | 0.035** |
| | (0.021) | (0.021) | (0.017) | (0.017) |
| Industry × Year FE | Yes | Yes | Yes | Yes |
| Occupation × Year FE | Yes | Yes | Yes | Yes |
| Year FE | No | Yes | No | Yes |
| Month FE | No | Yes | No | Yes |
| Month-Year FE | Yes | No | Yes | No |
| Metropolitan Area FE | Yes | Yes | Yes | Yes |
| Female × Metropolitan Area FE | Yes | Yes | Yes | Yes |
| Observations | 38,411 | 38,411 | 36,328 | 36,328 |
| Adjusted $R^2$ | 0.357 | 0.358 | 0.420 | 0.419 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Increases men's weekly earnings by 0.9% ($\approx \$9.11$/week), with an additional 6.6% ($\approx \$66.84$/week) increase for women.
- Increases men's hourly wage by 1.9% ($\approx \$0.47$/hour), with an additional 3.7% ($\approx \$0.92$/hour) increase for women.
- The positive marginal effects for women relative to men indicate a **reduction in the gender gap** following pay transparency.

# Event Study: Gender Earning/Wage Gap




Positive marginal effects for women indicate larger gains relative to men, meaning that the gender earnings and wage gaps decrease, particularly in the short run.

# Heterogeneous Effects on Gender Earnings/Wage Gaps by Education

| | ln(Weekly earnings) | | ln(Hourly wage) | |
|---|---|---|---|---|
| | Low educated | High educated | Low educated | High educated |
| Colorado × Post | 0.004 | 0.018 | 0.029 | 0.015 |
| | (0.022) | (0.024) | (0.031) | (0.023) |
| Colorado × Post × Female | -0.030 | 0.088*** | -0.043 | 0.062** |
| | (0.056) | (0.032) | (0.044) | (0.028) |
| Observations | 11,081 | 27,327 | 10,378 | 25,947 |
| Adjusted $R^2$ | 0.263 | 0.331 | 0.319 | 0.365 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Positive marginal effects for women relative to men are statistically significant only among highly educated workers, implying a decrease in the gender earnings and wage gaps, while the effects for less-educated workers are imprecisely estimated.

# Effects of Pay Transparency among Younger Workers (25-44)

| | ln(Weekly earnings) | ln(Hourly wage) | ln(Weekly earnings) | ln(Hourly wage) |
|---|---|---|---|---|
| CO×Post | 0.002 | 0.011 | -0.008 | 0.011 |
| | (0.021) | (0.016) | (0.024) | (0.018) |
| CO×Post×Female | | | 0.020 | -0.000 |
| | | | (0.022) | (0.020) |
| Industry × Year FE | Yes | Yes | Yes | Yes |
| Occupation × Year FE | Yes | Yes | Yes | Yes |
| Month-Year FE | Yes | Yes | Yes | Yes |
| Metropolitan Area FE | Yes | Yes | Yes | Yes |
| Observations | 43,337 | 41,093 | 43,337 | 41,093 |
| Adjusted $R^2$ | 0.355 | 0.436 | 0.355 | 0.436 |

Standard errors in parentheses.

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

# Wage Effects of Pay Transparency by Job Mobility

- Job mobility is an important source of wage growth and may contribute to gender wage differences (Topel and Ward, 1992; Moscarini and Postel-Vinay, 2016, 2018; Faberman et al., 2026).
- Pay transparency may increase wages through two channels:
  - **Job changers:** identify and move to better-paying outside opportunities.
  - **Job stayers:** use outside wage information to strengthen bargaining with current employers.
- **Question:** Do wage gains require workers to change jobs?

▸ Sum Job Mobility

# Wage Effects of Pay Transparency by Job Mobility

Job changers are defined as individuals who have ever changed employers or industries during the observed period.

| | Separate Samples | | Pooled Sample |
|---|---|---|---|
| | Job Changer | Job Stayer | Interaction |
| CO×Post | 0.062* | 0.034*** | 0.032*** |
| | (0.037) | (0.012) | (0.013) |
| CO×Post×Job Changer | | | 0.025 |
| | | | (0.018) |
| Observations | 7,575 | 28,749 | 36,328 |
| Adjusted $R^2$ | 0.417 | 0.422 | 0.421 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Hourly wages increase for both job changers and stayers, with larger gains among changers.

▸ Job Mobility

# Wage Gap Effects of Pay Transparency by Job Mobility

| | Changer | Stayer |
|---|---|---|
| CO × Post | 0.008 | 0.021 |
| | (0.051) | (0.014) |
| CO × Post × Female | 0.112** | 0.027* |
| | (0.048) | (0.014) |
| Observations | 7,572 | 28,749 |
| Adjusted $R^2$ | 0.413 | 0.422 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Larger positive marginal effects for women among job changers imply a larger decrease in the gender wage gap.
- Unlike Feng (2024), the gender wage gap also decreases significantly among job stayers.

# Heterogeneous Wage Effects across Occupations

- The value of information about outside wage opportunities may depend on the wage structure of workers' occupations.
- I examine heterogeneity along three pre-policy occupation characteristics:
  - **Wage dispersion:** differences in the range of potential wage opportunities within occupations.
  - **Wage level:** differences in average compensation across occupations.
  - **Age-wage profile:** differences in the relationship between age and wages across occupations.
- **Question:** Where is information about outside wage opportunities most valuable?

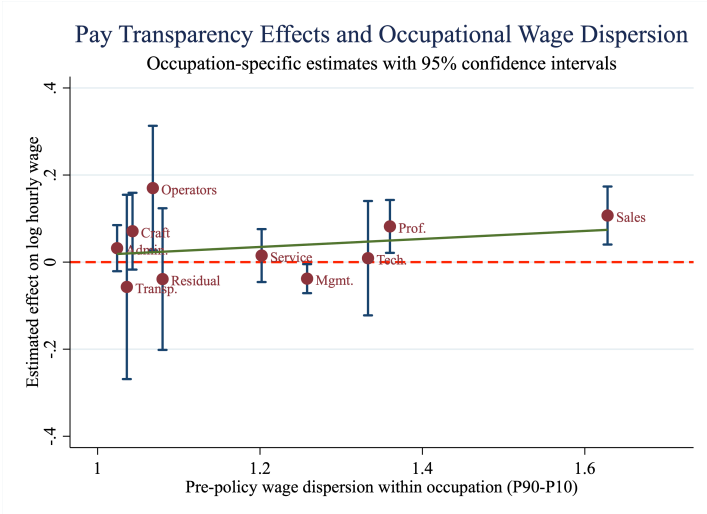
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# Wage Distribution by Occupations



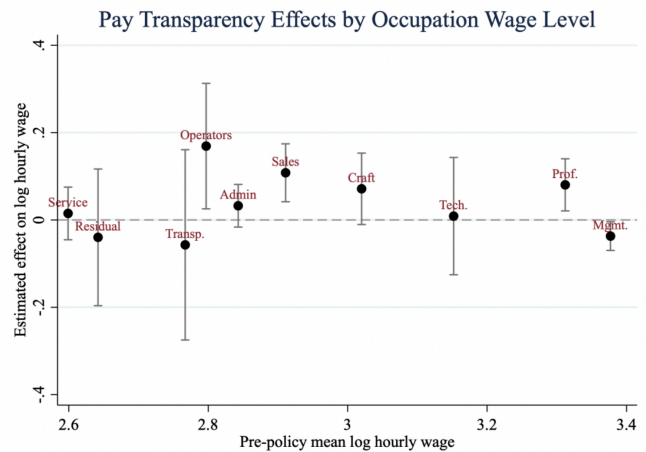

Sample includes employed workers aged 45--64. Protective service and service occupations are combined into Service.

# Heterogeneous Effects by Wage Dispersion




- Occupations with greater pre-policy wage dispersion tend to experience larger wage gains.
- Wage information may be more valuable when wages vary more widely.

# Heterogeneous Effects by Wage Level




- **Largest effects** in middle-wage occupations.
- **Some positive effects** in high-wage occupations.
- **Limited effects** in low-wage occupations.

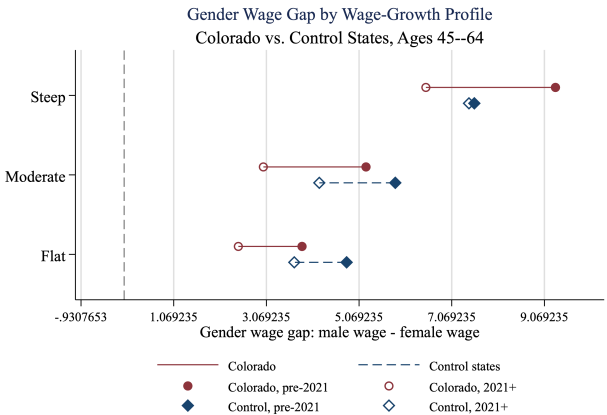
▸ Hetero Wage Level

# Heterogeneous Effects by Local Concentration

| | Low HHI | Medium HHI | High HHI | Pooled sample |
|---|---|---|---|---|
| *Panel A: Payroll-based HHI* | | | | |
| Colorado × Post | 0.037** | 0.041 | 0.053* | 0.032* |
| | (0.014) | (0.025) | (0.027) | (0.016) |
| Colorado × Post × Medium HHI | | | | 0.007 |
| | | | | (0.024) |
| Colorado × Post × High HHI | | | | 0.014 |
| | | | | (0.020) |
| Observations | 10,587 | 10,743 | 10,638 | 31,973 |
| Adjusted $R^2$ | 0.406 | 0.418 | 0.313 | 0.426 |
| *Panel B: Employment-based HHI* | | | | |
| Colorado × Post | 0.027* | 0.048** | 0.048* | 0.021 |
| | (0.015) | (0.021) | (0.025) | (0.014) |
| Colorado × Post × Medium HHI | | | | 0.029 |
| | | | | (0.026) |
| Colorado × Post × High HHI | | | | 0.022 |
| | | | | (0.019) |
| Observations | 10,741 | 10,465 | 10,762 | 31,973 |
| Adjusted $R^2$ | 0.403 | 0.427 | 0.308 | 0.426 |

▸ DefHHI

# Heterogeneous Effects by Wage Growth




Maroon solid lines: Colorado. Navy dashed lines: control states. Filled markers: pre-2021. Hollow markers: 2021+.

| | Steep | Moderate | Flat |
|---|---|---|---|
| CO×Post | -0.007 | 0.019 | 0.006 |
| | (0.026) | (0.023) | (0.041) |
| CO×Post×Female | 0.068** | 0.046** | 0.009 |
| | (0.025) | (0.022) | (0.092) |
| Observations | 17,660 | 14,824 | 3,827 |
| Adjusted $R^2$ | 0.251 | 0.318 | 0.226 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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# Effects of Pay Transparency on Employment and Working Hours

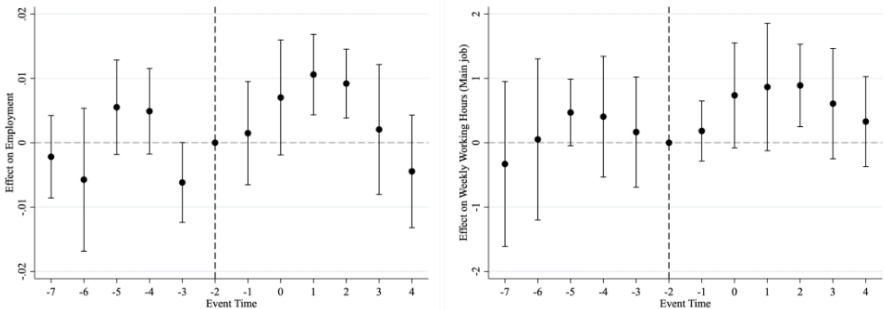
| | Employment | | Working hours: main job | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| CO × Post | 0.006*** | 0.008*** | 0.603*** | 0.536** |
| | (0.002) | (0.002) | (0.206) | (0.202) |
| Metropolitan Area FE | Yes | Yes | Yes | Yes |
| Year FE | No | Yes | No | Yes |
| Month FE | No | Yes | No | Yes |
| Year × Month FE | Yes | No | Yes | No |
| Industry × Year FE | Yes | Yes | Yes | Yes |
| Occupation × Year FE | Yes | Yes | Yes | Yes |
| Industry FE | No | No | No | No |
| Occupation FE | No | No | No | No |
| Observations | 219,360 | 219,360 | 143,174 | 143,174 |
| Adjusted $R^2$ | 0.871 | 0.871 | 0.119 | 0.119 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Employment increases by 0.6 p.p. (0.9% of the control mean).
- Conditional weekly hours increase by 0.60 hours (1.5% of the control mean).

# Event Study: Employment and Working Hours




- Both employment and working hours increase immediately after implementation, with significant effects in the short terms.
- Both effects diminish over time, suggesting primarily short-run labor supply responses.

# Workers' Beliefs about Outside Job Opportunities

- A key mechanism through which pay transparency may affect worker outcomes is by improving information about **outside job opportunities**.
- I use the **Survey of Consumer Expectations (SCE) Labor Market Survey** to examine whether Colorado's pay transparency law changed workers' beliefs about:
  - **Expected wages from outside job opportunities**
  - **Reservation wages**
  - **Perceived probability of receiving a job offer**

# SCE Labor Market Survey: Workers' Beliefs

**Data:** Federal Reserve Bank of New York's Survey of Consumer Expectations (SCE) Labor Market Survey.

- Provides direct measures of workers' expectations and beliefs about outside labor market opportunities.

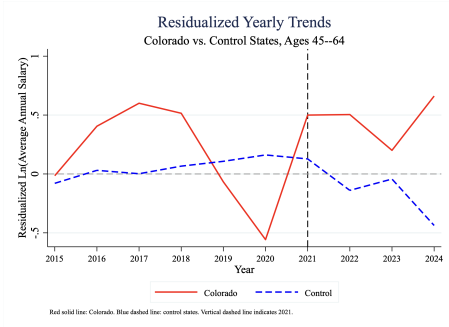
**Sample:** Workers aged 45–64, 2015–2024; exclude self-employed workers.

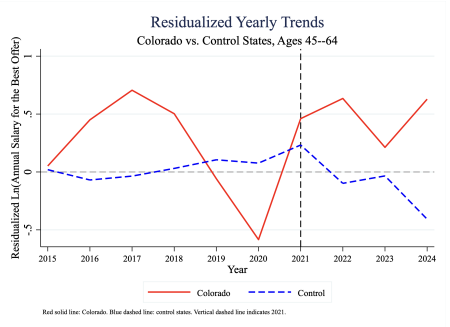
- Colorado vs. states without state- or city-level pay transparency laws.
- 7,813 observations: 242 Colorado and 7,571 control states.

**Belief measures:**

- **Outside wages:** expected salary and expected best-offer salary.
- **Reservation wage:** lowest wage willing to accept.
- **Job-finding prospects:** perceived probability of receiving at least one offer within four months.

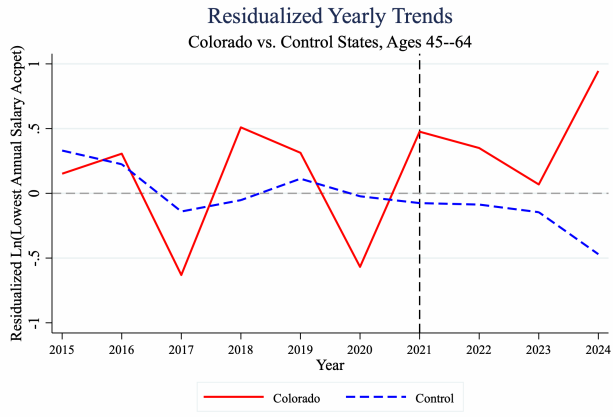
# Workers' Beliefs about Annual Salaries



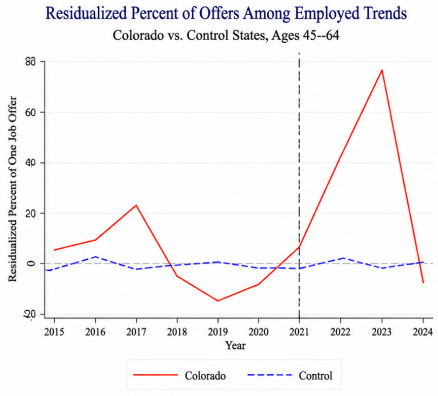

▸ Belief Wage

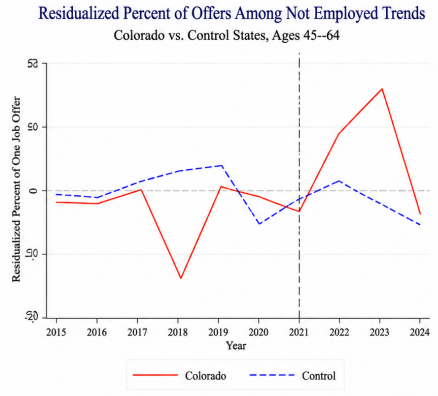
# Workers' Reservation Wage




▸ Belief Wage

# Workers' Beliefs about Percent of Offers




Red solid line: Colorado. Blue dashed line: control states. Vertical dashed line indicates 2021.

▸ Belief Offer

# Conclusion

- **Wages:** Pay transparency increases earnings and wages among mid- to late-career workers, particularly women and highly educated workers, and reduces the gender wage gap. Effects are limited among younger workers.
- **Who benefits**: Wage gains are larger among job changers and in occupations with greater wage dispersion, moderate age-wage profiles, and higher HHI. Gender-gap reductions are largest in occupations with steep age-wage profiles.
- **Labor supply:** Pay transparency increases employment and working hours, primarily in the short run.
- **Beliefs:** Workers revise upward their expected outside-option wages and reservation wages.
- **In sum:** Reducing compensation information frictions can improve workers' labor market outcomes and reduce gender inequality, and the benefits are heterogeneous.

# References I


Arnold, D., Quach, S., and Taska, B. (2025). The impact of pay transparency in job postings on the labor market. Technical report, National Bureau of Economic Research.

Blundell, R., French, E., and Tetlow, G. (2016). Retirement incentives and labor supply. In *Handbook of the Economics of Population Aging*. Elsevier.

Brown, M., Dwyer, R., Moulton, S., and Park, S. (2025). Who supports young workers through job loss? Working paper, The Ohio State University.

Faberman, R. J., Mueller, A. I., and Şahin, A. (2026). Job search, job amenities and the gender pay gap. Technical report, National Bureau of Economic Research.

Jäger, S., Roth, C., Roussille, N., and Schoefer, B. (2024). Worker beliefs about outside options. *The Quarterly Journal of Economics*, 139(3):1505–1556.

Krueger, A. B. and Mueller, A. I. (2016). A contribution to the empirics of reservation wages. *American Economic Journal: Economic Policy*, 8(1):142–179.

# References II


Maestas, N. and Li, X. (2006). Discouraged workers? job search outcomes of older workers. Working Paper WP 2006-133, Michigan Retirement Research Center.

Moscarini, G. and Postel-Vinay, F. (2016). Did the job ladder fail after the great recession? *American Economic Review*, 106(5):131–134.

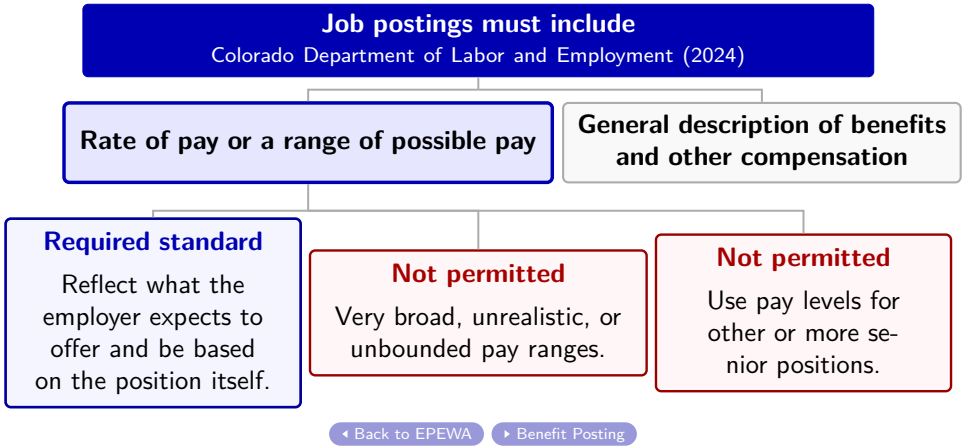
Moscarini, G. and Postel-Vinay, F. (2018). The job ladder: Inflation versus reallocation. *Annual Review of Economics*, 10:165–188.

Mueller, A. I., Spinnewijn, J., and Topa, G. (2021). Job seekers' perceptions and employment prospects: Heterogeneity, duration dependence, and bias. *American Economic Review*, 111(1):324–363.

Neumark, D., Burn, I., and Button, P. (2019). Is it harder for older workers to find jobs? new and improved evidence from a field experiment. *Journal of Political Economy*, 127(2):922–970.

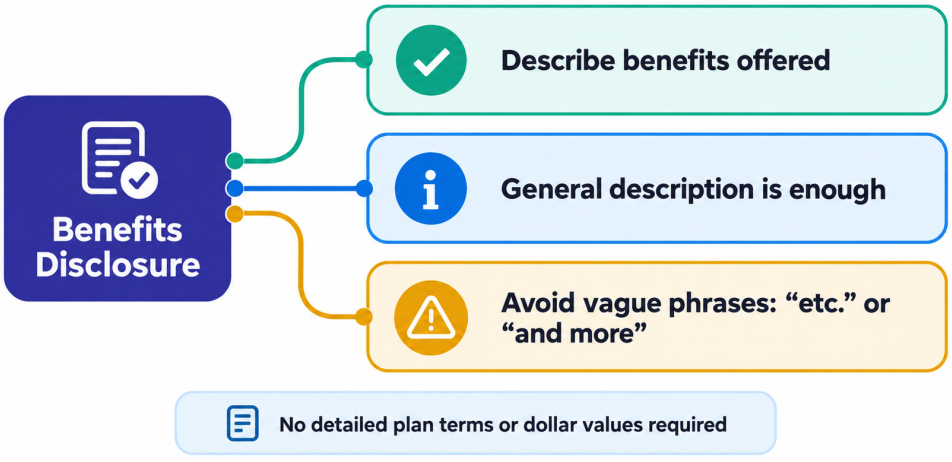
Topel, R. H. and Ward, M. P. (1992). Job mobility and the careers of young men. *The Quarterly Journal of Economics*, 107(2):439–479.

# Policy Background: Pay Posting Requirement




Back to EPEWA Benefit Posting

# Policy Background: Benefits Posting Requirement




Pay Posting

# Policy Background: Enforcement of EPEWA




▸ Policy Enforcement

# Why Focus on Colorado?

- **First statewide policy:** Colorado implemented EPEWA in January 2021, providing the longest post-treatment period in the sample.
- **Broad coverage:** Colorado's law applies to employers regardless of firm size, while California and Washington apply only to employers with at least 15 employees.
- **More comprehensive disclosure:** Colorado requires disclosure of pay ranges, benefits, and other compensation.
- **Cleaner policy setting:** Later reforms provide shorter post-treatment periods, and Washington introduced other major labor-market policies around the same time.

Sample Selection

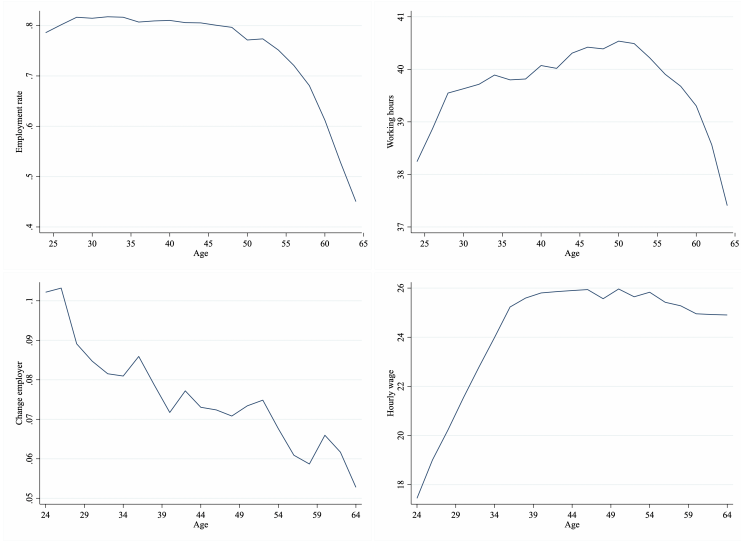
# Why Choose 45-64?

**Why ages 45–64?**

- U.S. age-discrimination law protects workers age **40 and older**.
- Retirement-related policies become more relevant in workers' **50s**.
- Age 64 keeps the sample below Medicare eligibility at 65.

Sample Selection

# Age Trends Panels



# Event Study Specifications: Wage and Labor Supply

$$Y_{igt} = \sum_{e=-7}^{4} \alpha_e \left(\mathrm{CO}_{ig} \times D_t^e\right) + X_{igt}'\gamma + \lambda M_{st} + \omega_{oy(t)} + \mu_{jy(t)} + \delta_t + \theta_g + \varepsilon_{igt}. \quad (3)$$

- **Event-time indicator:** The term $D_t^e \equiv \mathbf{1}\left\{\mathrm{y}(t) - 2021 = e\right\}$ equals one when calendar time $y(t)$ is $e$ periods from the implementation of Colorado's pay transparency policy, and zero otherwise.
- The coefficient $\alpha_e$ measures the dynamic policy effect at event time $e$, relative to the omitted pre-policy baseline period.
- Because year-month cells contain relatively few observations, I define event time at the annual level.

DID Main

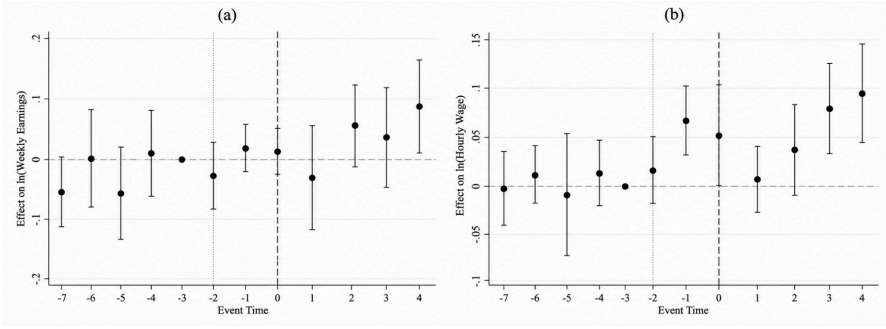
# Event Study Specifications: Gender Wage Gap (League and Shi, 2025)

$$Y_{igt} = \sum_{e=-7}^{4} \alpha_{1e}^{G} (\mathrm{CO}_{ig} \times D_t^e) + \sum_{e=-7}^{4} \alpha_{2e}^{G} (\mathrm{CO}_{ig} \times D_t^e \times \mathit{Female}_i) + \phi(\mathit{Post}_t \times \mathit{Female}_i) + X_{igt}'\gamma + \lambda M_{st} + \omega_{oy(t)} + \mu_{jy(t)} + \delta_t + \eta_{fg} + \varepsilon_{igt}. \tag{4}$$

- The coefficients $\alpha_{1e}^{G}$ measure the dynamic effects of pay transparency on men's hourly wages, relative to the omitted pre-policy baseline period.
- The coefficients $\alpha_{2e}^{G}$ measure how the policy effects on women's hourly wages differ from those on men's hourly wages.
- A positive $\alpha_{2e}^{G}$ indicates that the policy increases women's wages relative to men's wages, thereby decreasing the gender wage gap.

▸ DID Gap

# Event Study: Alternative Based year (2018)




▸ EsWage

# Heterogeneous Effects: Effects by Whether Hourly Paid

| | Not hourly paid | | Hourly paid | |
|---|---|---|---|---|
| | ln(Weekly earnings) | ln(Hourly wage) | ln(Weekly earnings) | ln(Hourly wage) |
| CO × Post | 0.043*** | 0.041*** | 0.053** | 0.024* |
| | (0.017) | (0.013) | (0.022) | (0.012) |
| Observations | 20,067 | 19,039 | 18,341 | 17,285 |
| Adjusted $R^2$ | 0.257 | 0.284 | 0.249 | 0.343 |

Standard errors clustered at the metropolitan-area level are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Main Result Wage

# Heterogeneous Effects on Gender Earnings/Wage Gaps by Education

| | ln(Weekly earnings) | | ln(Hourly wage) | |
|---|---|---|---|---|
| | Low educated | High educated | Low educated | High educated |
| Colorado × Post | 0.004 | 0.018 | 0.029 | 0.015 |
| | (0.022) | (0.024) | (0.031) | (0.023) |
| Colorado × Post × Female | -0.030 | 0.088*** | -0.043 | 0.062** |
| | (0.056) | (0.032) | (0.044) | (0.028) |
| Observations | 11,081 | 27,327 | 10,378 | 25,947 |
| Adjusted $R^2$ | 0.263 | 0.331 | 0.319 | 0.365 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Gender gap reductions are statistically significant only among highly educated workers, while estimates for less-educated workers are imprecise.

▸ Main Gap

# Sample Composition by Job Mobility Status

| | Stayer mean | Changer mean | Difference |
|---|---|---|---|
| Age | 53.849 | 53.777 | 0.073 |
| Male | 0.453 | 0.469 | -0.016** |
| Married | 0.681 | 0.658 | 0.023*** |
| Less high school | 0.037 | 0.045 | -0.008*** |
| High school | 0.262 | 0.262 | -0.000 |
| Some college | 0.149 | 0.158 | -0.009* |
| Associate degree | 0.112 | 0.114 | -0.001 |
| Bachelor degree | 0.257 | 0.264 | -0.006 |
| Master or more | 0.183 | 0.158 | 0.024*** |
| Hispanic | 0.050 | 0.057 | -0.007** |
| White | 0.815 | 0.762 | 0.053*** |
| Black | 0.095 | 0.137 | -0.042*** |
| Family size | 2.650 | 2.689 | -0.039** |
| Has child | 1.429 | 1.446 | -0.017 |
| Number of children | 0.766 | 0.773 | -0.007 |
| Hourly paid | 0.484 | 0.514 | -0.030*** |
| Public sector | 0.199 | 0.145 | 0.054*** |
| Main-job hours | 40.255 | 39.539 | 0.716*** |
| ln(Hourly wage) | 3.064 | 3.024 | 0.040*** |
| ln(Weekly earnings) | 6.719 | 6.645 | 0.074*** |

Difference is defined as stayer mean minus changer mean.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# Effect on Job Mobility

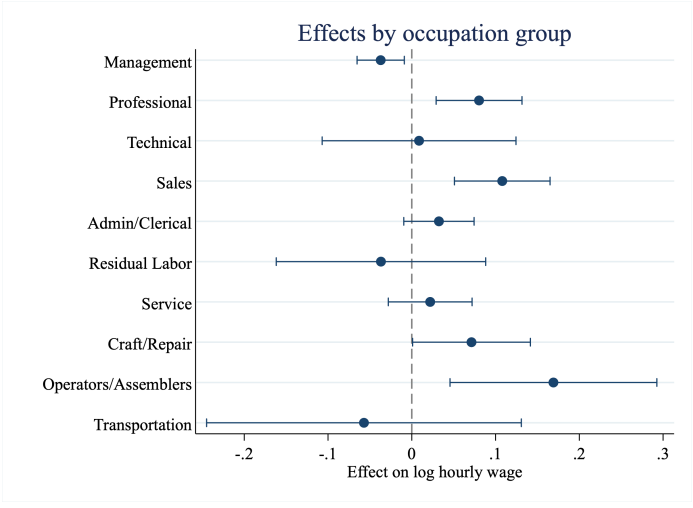
| | Ever Changed employer | Ever Changed occupation | Ever Changed industry | Change occupation | Change industry |
|---|---|---|---|---|---|
| CO×Post | -0.003 | 0.011 | 0.023 | 0.009 | 0.008 |
| | (0.010) | (0.020) | (0.017) | (0.007) | (0.006) |
| Observations | 143,425 | 151,623 | 151,623 | 121,427 | 121,427 |
| Adjusted $R^2$ | 0.011 | 0.045 | 0.056 | 0.017 | 0.017 |

Standard errors in parentheses

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

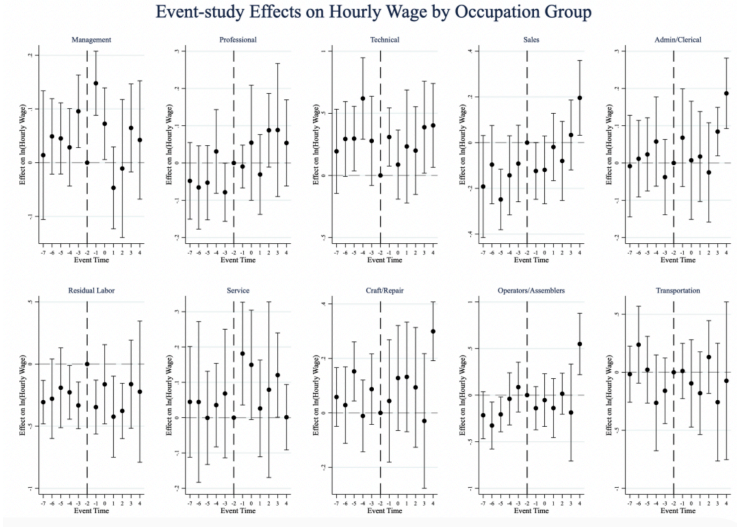
▸ Job Mobility Wage

# Heterogeneous Effects by Occupations

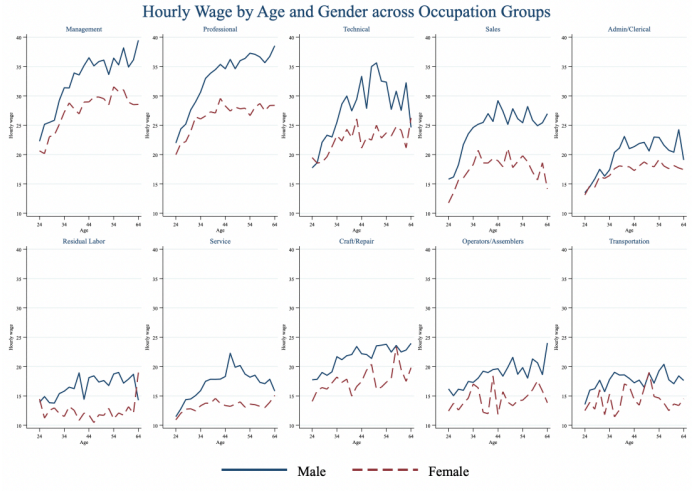



▸ Occupations

# Event Studies by Occupations



# Hourly Wage by Age and Gender across Occupations




▸ Gap Occ

# Heterogeneous effect of pay transparency based on occupation wage level

| | High | Median | Low |
|---|---|---|---|
| CO×Post | 0.027 | 0.053*** | 0.007 |
| | (0.019) | (0.018) | (0.025) |
| Observations | 17,663 | 12,447 | 6,213 |
| Adjusted $R^2$ | 0.250 | 0.270 | 0.264 |

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

▸ Wage Level

# Measuring Local Labor Market Concentration

- **Data:** 2018 employer concentration measures from Handwerker and Dey (2022).
- **Labor market definition:** Occupation $\times$ metropolitan area.

- **Payroll-based HHI:** Employer shares of total payroll.

$$HHI_{og}^{P} = \sum_{e \in \Omega_{og}} \left( \frac{Payroll_{eog}}{\sum_{j \in \Omega_{og}} Payroll_{jog}} \right)^2$$

- **Employment-based HHI:** Employer shares of total employment.

$$HHI_{og}^{E} = \sum_{e \in \Omega_{og}} \left( \frac{Employment_{eog}}{\sum_{j \in \Omega_{og}} Employment_{jog}} \right)^2$$

▸ HHI

# Effect on Workers' Belief about Wage

| | ln(Annual salary) | ln(Best annual salary) | ln(Reservation wage) |
|---|---|---|---|
| CO×Post | 0.468** | 0.404* | 0.478** |
| | (0.208) | (0.208) | (0.195) |
| Year FE | Yes | Yes | Yes |
| State FE | Yes | Yes | Yes |
| Observations | 4,766 | 4,678 | 7,496 |
| Adjusted $R^2$ | 0.127 | 0.136 | 0.144 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Belief Wage Trends

# Effect on Workers' Belief about Percent Offers

| | Probability of ≥1 Job Offer (Employed) | Probability of ≥1 Job Offer (Not employed) |
|---|---|---|
| CO×Post | 0.143 | 0.092** |
| | (0.096) | (0.043) |
| Year FE | Yes | Yes |
| State FE | Yes | Yes |
| Observations | 1,644 | 5,351 |
| Adjusted $R^2$ | 0.105 | 0.019 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Belief Offer Trends

# Robustness Check: Alternative Control Group

| | Employment | Main-job hours | ln(Weekly earnings) | ln(Hourly wage) | ln(Hourly wage) |
|---|---|---|---|---|---|
| CO×Post | 0.006*** | 0.404* | 0.048*** | 0.039*** | 0.006 |
| | (0.002) | (0.202) | (0.013) | (0.008) | (0.014) |
| CO×Post×Female | | | | | 0.066*** |
| | | | | | (0.015) |
| Observations | 1,618,921 | 1,055,906 | 280,799 | 266,469 | 266,465 |
| Adjusted $R^2$ | 0.876 | 0.111 | 0.344 | 0.415 | 0.415 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Main Result Wage

# Robustness Check: Effects of SHB on Wage

| | Employment | Main-job hours | ln(Weekly earnings) | ln(Hourly wage) |
|---|---|---|---|---|
| PT×SHB×Post | 0.006*** | 0.438** | 0.053*** | 0.044*** |
| | (0.002) | (0.212) | (0.014) | (0.009) |
| SHB×Post | 0.001 | -0.052 | -0.005 | -0.004 |
| | (0.001) | (0.088) | (0.007) | (0.006) |
| Observations | 1,618,921 | 1,056,498 | 280,799 | 266,469 |
| Adjusted $R^2$ | 0.876 | 0.111 | 0.344 | 0.415 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

▸ Main Result Wage

# Robustness Check: Effects of SHB on Wage Gap

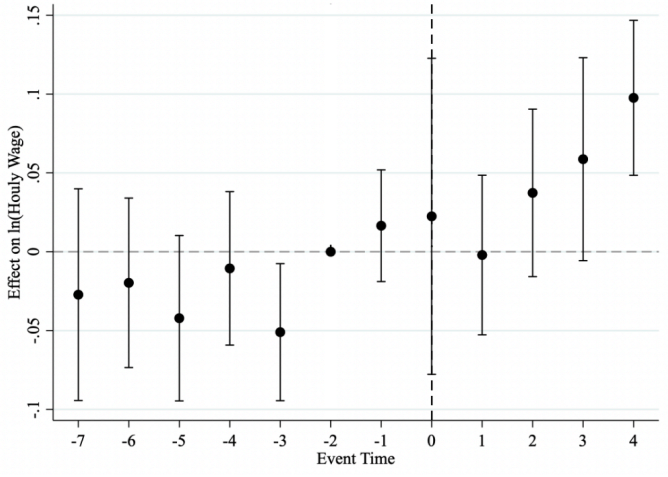
| | ln(Hourly wage) |
|---|---|
| PT×SHB×Post | 0.019 |
| | (0.016) |
| PT×SHB×Post×Female | 0.051*** |
| | (0.018) |
| SHB×Post | -0.014* |
| | (0.007) |
| SHB×Post×Female | 0.016 |
| | (0.011) |
| Observations | 266,469 |
| Adjusted $R^2$ | 0.415 |

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Main Result Wage

# Diagnosing the 2020 wage fluctuation




▸ EsWage

# Diagnosing the 2020 wage fluctuation

